

TKF exercises

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1 Instantaneous model

Consider the Thorne, Kishino, and Felsenstein 1992 model [3], known as TKF92:

- A biological sequence (DNA, RNA, or protein) consists of an *immortal link* followed by zero or more *mortal links*, into which the characters of the sequence (if it is nonempty) are partitioned;
- New links are inserted immediately to the right of existing (parent) links at a rate λ per parent link (parent links can be mortal or immortal; child links are always mortal);
- Mortal links are deleted at a rate $\mu > \lambda$ per mortal link;
- Each mortal link has associated with it n residues, where n is independently drawn for each link from a geometric distribution with support $1, 2, 3 \dots$ and extension probability parameter ρ ;
- Each residue is drawn (on insertion) from a distribution π , and experiences substitution according to a finite-state continuous-time Markov chain with transition rate matrix Q , for which π is the equilibrium distribution (so $\pi Q = 0$).

Questions about this model:

1. Consider the outer birth-death process describing the evolution of the number of mortal links over time. Let $r_n(t)$ be the probability that, at time t , there are n mortal links. Write down a series of coupled first-order ODEs for the $r_n(t)$.
2. Does this outer process admit an equilibrium distribution over the number of links? If so, what constraints specified in the model description ensure this is the case? If not, what constraints need to be added?
3. Assuming appropriate constraints that ensure the equilibrium exists, derive $\rho_n = \lim_{t \rightarrow \infty} r_n(t)$, the probability that, at equilibrium, there are n mortal links.

4. What is (i) the expected number of mortal links, at equilibrium; (ii) the probability that, when sampled at equilibrium, the system has zero links?
5. What is the expected number of residues per mortal link?
6. What is the expected total number of residues, at equilibrium?
7. Using your answers to previous questions, describe a Hidden Markov Model (state space, transition probabilities, and emission probabilities) whose output distribution over sequences is the TKF92 equilibrium distribution.
8. Is there a one-to-one mapping between states of the TKF92 model, and sequences over the character alphabet? If not, which has “more” information? What does this information correspond to?

2 Finite-time solution

Consider the following time-varying probabilities, defined for all $n \geq 0$:

- $p_n(t)$, the probability that (after time t) a mortal link has survived, with n surviving children born during the time interval $[0, t]$;
- $q_n(t)$, the probability that (after time t) a mortal link has died, leaving n orphaned children born during the time interval $[0, t]$;
- $r_n(t)$, the probability that (after time t) the immortal link has n surviving children born during the time interval $[0, t]$.

Questions:

1. Write down coupled first-order ODEs relating these time-varying probabilities. Refer to the TKF papers [2, 3] to confirm your results and/or if you get stuck.
2. Is the $r_n(t)$ from this section the same as $r_n(t)$ from Section 1? What, if anything, is different?
3. The process whose state vector is given by $r_n(t)$ is a linear birth-death process with constant immigration. What does the “immigration” part refer to? What is the immigration rate?
4. Use the following trial solution to obtain ODEs for $\alpha(t)$, $\beta(t)$, and $\gamma(t)$. Solve these ODEs, using integrating factors or otherwise.

$$\begin{aligned}
 p_n(t) &= \alpha \beta^n (1 - \beta) \\
 q_0(t) &= (1 - \alpha)(1 - \gamma) \\
 q_{n>0}(t) &= (1 - \alpha) \gamma \beta^{n-1} (1 - \beta) \\
 r_n(t) &= \beta^n (1 - \beta)
 \end{aligned}$$

5. Confirm that your solutions are consistent with those given in the TKF91 paper [2]. What general form do these solutions take, in terms of their dependency on n ?
6. (Advanced) Feller [1] suggests the following systematic approach to solving these ODEs, e.g. for $r_n(t)$:
 - (a) construct a generating function $g(s, t)$, by multiplying by s^n and summing over n ;
 - (b) convert the differential operators on r_n , and other operations such as multiplying by n , to operations on g involving partial derivatives;
 - (c) thus obtain a PDE for g ;
 - (d) solve this PDE using the method of characteristics.

Alternatively, Karlin and McGregor show that the solution can be obtained as an expansion of Meixner polynomials, which are eigenfunctions of the underlying differential operator. Which of these two approaches do you find more intuitive? Take them as far as you can. [4].

3 Hidden Markov Model

Describe an HMM that generates pairwise alignments from the finite-time joint distribution of ancestor and descendant sequence under the TKF92 model, assuming the ancestral sequence is sampled from the equilibrium distribution.

Your model should have

- Five states: S(tart), E(nd), M(atch), I(nsert), D(elete).
- Hence a 5×5 transition probability matrix. What is it? (How do you choose to handle transitions from the End state?)
- The Match state emits aligned ancestor-descendant character pairs. What is the emission probability of the pair (x, y) ?
- The Insert state emits unaligned descendant characters (ϵ, y) and the Delete state emits unaligned ancestral characters (x, ϵ) , where ϵ represents a gap. What distribution(s) are these characters $(x$ and $y)$ drawn from?

Confirm a few expected properties of your HMM:

1. that it is appropriately normalized (probabilities of all transitions exiting each state must sum to one; probabilities of emissions from any state must sum to one);
2. that it has no null cycles (transition paths starting and ending in the same state that have finite probability and no associated emissions).

4 HMM algorithms

Let X denote an ancestral sequence, Y a descendant sequence, A their alignment, t the evolutionary time separating them, and $\theta = (\lambda, \mu, \rho, Q, \pi)$ the TKF92 model parameters.

Describe each of these tasks in words, and (as appropriate) give examples of a context in which each of them might come up. Implement at least two of them, and outline how you might implement the others.

1. Calculate $P(A, X, Y|\theta, t)$
2. Find $\hat{t} = \operatorname{argmax}_t P(A, X, Y|\theta, t)$
3. Calculate $P(X, Y|\theta, t)$
4. Calculate $P(Y|X, \theta, t)$
5. Find $\hat{A} = \operatorname{argmax}_A P(A|X, Y, \theta, t)$
6. Find $\hat{\theta} = \operatorname{argmax}_{\theta} P(A, X, Y|\theta, t)$

References

- [1] William Feller. *An Introduction to Probability Theory and Its Applications*, volume 1. Wiley, January 1968.
- [2] Jeffrey L. Thorne, Hirohisa Kishino, and Joseph Felsenstein. An evolutionary model for maximum likelihood alignment of DNA sequences. *Journal of Molecular Evolution*, 33(2):114–124, August 1991.
- [3] Jeffrey L. Thorne, Hirohisa Kishino, and Joseph Felsenstein. Inching toward reality: An improved likelihood model of sequence evolution. *Journal of Molecular Evolution*, 34(1):3–16, January 1992.
- [4] Erik A. van Doorn. Birth–death processes and associated polynomials. *Journal of Computational and Applied Mathematics*, 153(1):497–506, 2003. Proceedings of the 6th International Symposium on Orthogonal Polynomials, Special Functions and their Applications, Rome, Italy, 18–22 June 2001.